

Set Theory Cheat Sheet: Days 1-5

A **set** is a collection of distinct objects, called its elements.

- We write $x \in A$ to mean that x is an element of the set A .
- We write $A \subseteq B$ to mean that A is a subset of B . This means every element of A is also an element of B .

Permanent names for sets:

$\emptyset$: The **empty set** is the unique set with no elements.

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$: The **natural numbers**, **integers**, **rational numbers**, and **real numbers** respectively.

Operations on sets:

- Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.
- Intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.
- Cartesian product: $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$.
- Power set: $\mathcal{P}(A) = \{X \mid X \subseteq A\}$.

Axioms for sets (ZFC):

Extensionality. Two sets are the same if they have the same elements.

Infinity. There exists an infinite set.

Pairing. If A and B are sets, then $\{A, B\}$ is a set.

Power set. If A is a set, then $\mathcal{P}(A)$ is a set.

Union. The union of a family of sets is a set.

Foundation. Every non-empty set A has an element B which is disjoint from A .

Replacement. If f is a function, then the range of f is a set.

Separation. If A is a set and P is a property, then $\{x \in A \mid P(x)\}$ is a set.

Choice. The cartesian product of a non-empty family of nonempty sets is nonempty.

Two sets have the same **cardinality** if there is a one-to-one correspondence between them mapping elements of one set to elements of the other.

Permanent names for cardinalities:

- $\aleph_0$: "Countable infinity". The cardinality of the set of natural numbers.
- $\mathfrak{c}$ or $2^{\aleph_0}$: "Continuum". The cardinality of the set of real numbers.